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|---|-----|-------|--|----------------|-----|
| 6 | (a) | (i) | either $P = V^2 / R$ or $P = VI$ <u>and</u> $V = IR$
$R = 4.0 \, \Omega$ | C1
A1 | [2] |
| | | (ii) | sketch vertical axis labelled appropriately
(straight) line from origin then curved in correct direction
line passes through 12 V, 3.0 A | B1
B1
B1 | [3] |
| | (b) | (i) | 2.0 kW | A1 | [1] |
| | | (ii) | 0.5 kW | A1 | [1] |
| | | (iii) | total resistance = $3R / 2$
power = 0.67 kW | C1
A1 | [2] |

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7 (a) either different forms of some element